

A Taste of Categorical Neuroscience

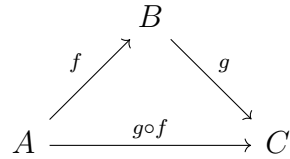
In the summer of 2017, Thomas Garrity, professor of mathematics at Williams College, began his lecture, *On Mathematical Maturity*, at the Park City Mathematics Institute by writing the following on the blackboard:

Functions describe the world.

Indeed functions are ubiquitous in mathematics, and thus also the world. However, in different disciplines, functions often look different. For example, in topology, the functions of interest are continuous functions; in algebra, the important functions are those that preserve the operations of addition and/or multiplication; and in computer science, the only functions that are relevant are those that are computable (i.e. can be written in the form of a computer program). Despite the differences between such classes of functions, there are many similarities. In 1945, Samuel Eilenberg and Saunders Mac Lane gathered the underlying properties that are essential to all contexts that functions appear in. In their paper, *General Theory of Natural Equivalences*, they call a network of “functions” with these properties a **category**.

I want to now give a brief glimpse into the ideas of category theory. Our starting point is the idea that every function has a **domain**, let’s call it A , and a **target**, which we’ll call B . A function with domain A and target B takes elements of A and maps them over to B . For example, if A is the surface of the Earth, and B is the real numbers, then the temperature is such a function that assigns to each place on Earth, the temperature there. We write such a function as $f : A \rightarrow B$, or simply $A \rightarrow B$. Notice how in any context, functions can be represented by

arrows in this way. The main idea of category theory is to view these arrows as the fundamental things you are interested in. A category is a network of such arrows with an algebraic structure that resembles composition of functions, i.e. the ability to apply one function after another. This operation can be viewed as in the following diagram.



In the diagram above, the arrow marked $g \circ f$ is the one obtained by applying f first, and then applying g to the result. Notice that this composition can be done when the target of f and the domain of g are the same. The takeaway from our dip into category theory is this: a **category** is a context which consists of **objects** we’re interested in studying, and **morphisms** (the arrows) which are processes that take these objects to other such objects. Whenever two morphisms meet at an object (“tip to tail”), they can be combined into a single morphism. The focus of category theory is that the structure of the network of morphisms is where the real information of a given discipline is contained. Ideally, I would like to give examples that demonstrate the power of category theory, but due to space limitations, let’s jump into neuroscience.

Kathryn Hess is an algebraic topologist who has recently started to investigate applications of algebraic topology and category theory to neuroscience. I will focus on her point of view as presented in two talks: *Towards a categorical approach to neuroscience* at the 2018 Applied Category Theory workshop, and her TEDx talk, *Digital Neuroscientist of the Future*. Her main approach is to construct a category where the objects are neurons and the morphisms are (reliable) connections between neurons. In order to explain this construction, let’s look at Hess’s picture of a neuron. A neuron has three main parts: the **soma** (cell body), incoming wires called **dendrites**, and outgoing wires called **axons**. She draws this as in Figure 1. The **synapses**, which are represented by yellow dots in the figure, are locations at which an axon of one neuron can connect to a dendrite of another neuron. Such a connection makes it possible for a neuron

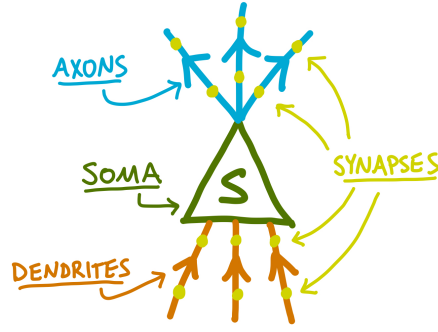


Figure 1: A neuron.

to communicate with another neuron. Note that these connections at a synapses are *directed*, i.e. the communication only flows in one direction, from the axon to the dendrite.

Now that we have defined our objects, let's define the morphisms. A first guess would be to define a morphism $f : A \rightarrow B$ to be any connection from an axon of neuron A to a dendrite of neuron B . Hess points out that this definition would not be too useful for the following reason. Firings of neurons are highly random, so a single connection between two neurons does not realistically form a reliable connection. Thus, we require approximately 6-7 such connections to form a morphism in our category. The domain (in the categorical sense above) of such a morphism

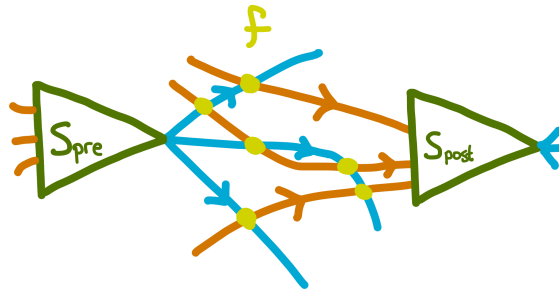


Figure 2: A morphism in the category of neurons.

is called the **pre-synaptic** neuron, and the target is called the **post-synaptic** neuron. Thus, a morphism $f : S_{pre} \rightarrow S_{post}$ in our category looks something like Figure 2.

Our last order of business in order to establish that this network of neurons is indeed a

category is to define composition of morphisms. In order to do this, note that we need to broaden what we call a morphism. Indeed if neuron A is connected to neuron B at 6-7 synapses, and B is connected to neuron C at 6-7 synapses, there is no reason to believe that A will also be connected to C at 6-7 more. Even if this was the case, such a connection could not be naturally interpreted as a composite of the first two connections. Thus, we broaden the term morphism in this context to mean a *path* of reliable connections of neurons. If we have connections as follows,

$$A \xrightarrow{f} B \xrightarrow{g} C,$$

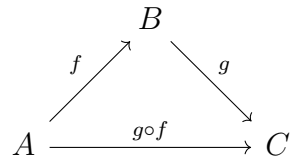
then we observe that A is indeed connected to C through f and g , in the sense that the neuron A can pass information to C by passing it along f to B , which then B passes to C along g . This process defines composition of morphisms which makes our neural network into a category.

Now, it is time for us to explore how Hess suggests such a category can be useful. While Hess acknowledges that the possibilities are great in number, for the sake of this essay I want to focus on two points. Both relate to the problem of the sheer magnitude of the number of neurons, and connections between them, in our brain. To get a glimpse of the issue, the number of synapses in a human brain is on the order of 100 trillion. This demonstrates that the network of neurons in our brain is unimaginably large and complex, so any detailed analysis becomes extremely difficult very quickly.

One way Hess suggests category theory can potentially help with this problem is the notion of **equivalence of categories** (Hess does not use this language; I am using more standard categorical terminology). Despite its name, equivalence of two categories does not mean they are the same. Equivalence roughly means that the structure of the morphisms in two categories is the same, even if one category is much larger than the other. In our context, equivalence of categories can be used to replace the large category of neurons with an equivalent smaller one, if such a category can be found. Studying this equivalent category would actually tell us

everything we need to know about the category of neurons, as far as the structure of connections is concerned. We'd only be missing the notion of scale, but we already know this.

The other application I want to address is one that Hess herself has worked on, and this involves a concept in topology called a **simplicial complex**, which is a geometric entity built by piecing together segments, triangles, and their higher-dimensional analogues. Here, view a segment as a connection between neurons, and view a triangle as a composition of two such connections. The diagram we used before



demonstrates how we can view this as a triangle. As we do this to longer strings of connections, we get the higher-dimensional analogues of triangles. Algebraic topology provides a large variety of tools of distinguishing such simplicial complexes, but these tools only reference the relevant *geometric* structure. Since the flow of information is directional among these connections, the category more accurately reflects the dynamic structure of the network.

In conclusion, I think Hess's work is a fascinating demonstration of the power of abstract mathematical constructions even in applied settings. In one respect, we saw how the communications of neurons can be represented geometrically by simplicial complexes, which then allows us to use the vast amounts of topology developed to study neuroscience. But more broadly, we see how category theory, initially motivated by the ubiquity of functions, can be relevant even in contexts where the morphisms are not functions, but directed connections. I believe that this perspective will be important for the future of neuroscience, and for other disciplines as well. Indeed, maybe I would rephrase Thomas Garrity's motto as:

Categories describe the world.